

Theory Assignment - 4

1. Cipher Text: H L D T Q I H J O W X Q C M A G

Key: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ find plain textDeterminant $|K| = (3 \times 7) - (2 \times 5) = 21 - 10 = 11$ inverse
Det of determinant

$$11^{-1} \text{ mod } 26$$

$$26 = 11(2) + 4$$

$$[det K]^{-1} adj[K]$$

$$11 = 4(2) + 3$$

$$= 11 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3(1)$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3(26) - 11(6) - 11$$

$$1 = 3(26) - 11(7)$$

$$H L D T Q I H J O W X Q C M A G$$

$$7 \ 11 \ 3 \ 14$$

$$\Rightarrow 26 - 7 = 19$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(7) + (14 \times 11) \\ (9 \times 7) + (5 \times 11) \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(3) + (14 \times 19) \\ 9(3) + (5 \times 19) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{bmatrix} P \\ S \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} (3 \times 16) + (14 \times 8) \\ (9 \times 16) + (5 \times 8) \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} F \\ C \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 7 + 14 \times 4 \\ 9 \times 7 + 5 \times 4 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{bmatrix} R \\ F \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 3 + 14 \times 23 \\ 9 \times 3 + 5 \times 23 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{bmatrix} T \\ M \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 22 + 14 \times 16 \\ 9 \times 22 + 5 \times 16 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 290 \\ 271 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{bmatrix} F \\ S \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 2 + 14 \times 12 \\ 9 \times 2 + 5 \times 12 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 6 \end{bmatrix} = \begin{bmatrix} S \\ A \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 0 + 14 \times 6 \\ 9 \times 0 + 5 \times 6 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} L \\ P \end{bmatrix}$$

PT $\Rightarrow$ TOP SECRET MESSAGE

2) Consider the Hill cipher Key Matrix
 $K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ Cipher Text: RXADQY Find Plain text

$$\Rightarrow \det \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241$$

$$241 \pmod{26} = 7$$

$$7^{-1} \Rightarrow 26 = 7(3) + 5 \Rightarrow 5 = 26 - 7(3)$$

$$7 = 5(1) + 2 \Rightarrow 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 26(5)$$

$$\Rightarrow 26 - 11 = 15$$

$$15 \begin{bmatrix} 14 & 19 \\ 5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 285 \\ -75 & 370 \end{bmatrix}$$

$$\pmod{26}$$

$$= \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$i) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 22 \times 23 \end{bmatrix} = \begin{bmatrix} 547 \\ 557 \end{bmatrix} \pmod{26} \\ = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$ii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 \times 0 + 1 \times 14 \\ 3 \times 0 + 22 \times 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26} \\ = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$iii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \times 3 + 1 \times 24 \\ 3 \times 3 + 22 \times 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 531 \end{bmatrix} \pmod{26} \\ = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{bmatrix} E \\ R \end{bmatrix}$$

PT = FLOWER

3 Relation b/w substitution, confusion, transposition & Diffusion.

→ i) Substitution & Confusion

Substitution replaces one character with another character.

Eg.: HELLO → LKMOOR

The relationship is:

Substitution creates confusion by hiding the relationship b/w original plain text character & ciphertext character.

Substitution → Confusion

makes it difficult to understand how the key & plaintext are related to the cipher text.

ii) Transposition → Diffusion

Transposition changes the position of character changing the character themselves.

Eg.: HELLO → LHOEL

The relationship

Transposition creates diffusion by spreading the influence of plaintext character across different positions in cipher text.

Transposition $\rightarrow$ Diffusion.

Diffusion spreads info of plaintext throughout the cipher text making pattern harder to recognize.

Cipher	Does it change the character / symbol value?	Does it change character position	Confusion / Diffusion	Justification
Shift cipher	Yes	No	Confusion	Replaces each character with a different character based on an addition key, obscuring the PT without moving character location.
Multiplicative cipher	Yes	No	Confusion	Replaces each character with another character using a mathematical multiplication modulus 26, creating substitution without moving position.

Affine Cipher	Yes	No	Confusion	Combines both shift & multiplicative principles to substitute characters, intensify focusing on altering the symbols value to hide relationship
Transposition Cipher	No	Yes	Diffusion	Keeps all original characters but shuffles them to different positions, spreading the linguistic pattern across the cipher text.

TranspositionEncryption

Encrypt the message = The arrow is pointing north direction "very".

- Simple

With Key = 4

- keyword

With Key = ATLAS

- double

Key 1 = BREAK

Key 2 = Log

i) Simple (Key = 4)

1	2	3	4
T	H	E	A
R	R	O	W
I	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
I	R	E	C
T	I	O	N

Cipher Text :-

TRZINRIT HRSNHTRI
EUPTNHEU AWOIUDLN

ii) Keyword (Key = ATLAS)

① ⑤ ③ ② ④

A T L A S

T H E A R

R O W I S

P O I N T

I N L N O

R T H D I

R E C T I

O N X X X

Cipher Text :-

TRPIRRO AINNDT

EWILHCK RSTOIX

HOONTEN

iii) Double Transposition (Key 1 = BREAK, Key 2 = Log)

② ⑤ ③ ① ④

B R E A K

T H E A R

R O W I S

P O I N T

I N L N O

R T H D I

R E C T I

O N X X X

Cipher text :-

AINNDT TRPIRRO

EWILHCK RSTOIX

HOONTEN

Log

A I N

N O T

X T R

P I R

R O E

W I L

H C X

R S T

O I I

X H O

O N T

E N X

Cipher text :-

ANXPRWHROXOE

IOTIOLIS IHNN

NTRREGLX7IOTX

Decryption :-

i) T I O U E M R N M E R E U G O S A N O W R S L T P
 E E E D O S S S N U A H T U B B I E M P L O D A E
 P E Z X O E L N S A

→ Simple With Key = 10

$$R = \frac{60}{10} = 6$$

R1	T	R	O	U	P	S	H	E	A	D
R2	I	N	G	W	E	S	T	N	E	E
R3	D	M	O	R	E	S	U	P	P	L
R4	I	E	S	S	E	N	O	G	E	N
R5	E	R	A	L	D	U	B	O	I	S
R6	M	E	N	T	O	A	Z	D	X	X

3 TROOPS HEADING WEST NEED MORE SUPPLIES
 SEND GENERAL DUBOIS MENT TO AID

ii) Keyword = SECRET

(T = I L E D T N H P T R S U I L T M A X S A X E R A T
 A N O W I S N E O T

$$R = \frac{36}{6} = 6$$

R1	I	H	I	S	T	S	R1	T	H	I	S	I	S
R2	L	P	I	A	A	N	R2	A	P	L	A	I	N
R3	E	T	T	X	N	R	R3	N	T	E	X	T	F
R4	O	R	M	E	O	O	R4	O	R	O	E	M	O
R5	T	S	A	R	N	T	R5	N	S	T	R	A	T
R6	N	O	X	X	I	X	R6	Z	O	N	X	X	X

THIS

THIS

THIS IS A PLAIN TEXT FOR DEMONSTRATION

iii) Keyword - PLAIN

CT = VEY EAX ARA ATX HGD

$$R = \frac{15}{5} = 3$$

⑤ ③ ① ④ ②

P L A I N

PT = HAVE A GREAT DAY

R1 H A V E A

R2 G R E A T

R3 D A X X X

Permutation:

Find the inverse permutation for

1) 4 3 1 5 2

: 1 2 3 4 5 → INDEX

4 3 1 5 2

Inverse = 3 5 2 1 4

2) 7 4 1 3 5 2 6

1 2 3 4 5 6 7 → INDEX

7 4 1 3 5 2 6

Inverse = 3 6 4 2 5 7 1

3) 4 1 3 2

1 2 3 4 → INDEX

4 1 3 2

Inverse = 2 4 3 1

4) 3 5 1 4 2

1 2 3 4 5 → INDEX

3 5 1 4 2

Inverse = 3 5 1 4 2